

tmx wizard + session-password redesign — design spec

Revision: 1 **Last modified:** 2026-07-05T00:00:00Z **Status:** approved (brainstorming phase) — pending implementation

Forensic anchor — verbatim user mandate (2026-07-05)

"1. When asked for name of new session at the end of name it must be appended random 4 number suffix. ... 2. When we are entering passwords, they **MUST** not be visible to a naked eye but presented with "*" characters as we are typing! 3. When we open sessions which are password protected, and we enter valid password, we are then asked twice to enter (maybe new) password! ... We **MUST BE** asked twice to enter password if we create new session password protected - the password and confirmation. 4. When user presses enter, so he does not want to create new session, he should have a choice of choosing one of existing sessions, or the option of none, to leave wizard..."

Root cause of requirement #3 (the double-password-prompt bug)

The interactive wizard (scripts/tmx-shell-init.sh.template) runs:

```
tmx attach -t "$1" 2>/dev/null || exec tmx new -s "$1"
```

The attach verb (scripts/tmx.template) checks for a persisted password **before** checking whether a live tmux session actually exists. For a session that was idle-recycled (killed by tmx-recycler.sh for inactivity, but whose password/state persists in tmx-state-bin's JSON store), this means: the operator is prompted to *verify* the (still-valid) password, succeeds, then the wrapper's final exec "\$TMUX_BIN" ... attach -t "\$NAME" fails because the socket is dead. That failure propagates back to the wizard's ||, which falls through to tmx new -s "\$1" — and the new verb *unconditionally* re-prompts "Enter password for session NAME

(blank = none)" and overwrites the stored hash with whatever is typed. This is not a real confirmation dialog; it is an accidental password-reset triggered by a stale-session attach failure.

Scope decisions (resolved via brainstorming Q&A, 2026-07-05)

1. **Suffix scope:** applies to **both** the interactive wizard and direct `tmx new -s NAME` CLI usage — but the wrapper (`tmx.template`) itself never invents the suffix; only the wizard (`tmx-shell-init.sh.template`) generates and appends it before calling `tmx new -s <name>-<suffix>`. This keeps `tmx new -s NAME` byte-exact for scripting/tests by construction, while still satisfying "both" — the wizard is the only caller that ever adds anything, and it does so unconditionally by default.
2. **Suffix opt-out:** `TMX_EXACT_NAME=1` (env var), read only by the wizard. When set, the wizard uses the typed name literally, still going through the always-create path (no attach-first probe).
3. **Existing-session picker UX:** numbered list (1) `name1`, 2) `name2`, ...) + 0) None (leave, bare shell). Zero existing sessions → skip the menu, go straight to bare shell (unchanged current default). Invalid selection → one re-prompt, then bare shell.
4. **Password confirmation mismatch:** retry up to 3 times; on the 3rd failure, abort — if the tmux session was already created in this invocation, tear it down (fail-closed: no half-configured session left behind).

Architecture

Components touched

Component	Change
<code>scripts/tmx-state/state.go + main.go</code>	New <code>has-password <session></code> subcommand: exit 0 = has password, 1 = record exists/no password, 2 = no record. Additive only.
<code>scripts/tmx.template</code>	New <code>_read_password_masked()</code> helper (bash, *-echo, backspace support). new verb restructured: password decision (verify-existing OR collect-new-with-confirm) moves before tmux session creation. attach verb restructured: has-session check moves before the password guard.
<code>scripts/tmx-shell-init.sh.template</code>	Replace "Enter session name" → attach-else-new chain with: name typed → generate 4-digit suffix (POSIX

Component	Change
	awk) → always exec tmx new -s "name-NNNN"; blank → numbered existing-session picker → attach chosen or bare shell.
scripts/tests/ 68_session_lifecycle.sh	Rewrite C6 (recreate-after-recycle: now verify-once, correct-accepts/wrong-rejects, no re-set) and C7 (post-delete recreate: now set+confirm-twice) per §11.4.120 gate reconciliation.
scripts/tests/ 66_session_password.sh	Add has-password exit-code cases (0/1/2).
New tests 77-84	See test plan below.
Docs	README, docs/guides/tmx-shell-integration.md, new docs/guides/tmx-session-passwords.md, FAQ, diagrams, workable-items DB entries, CONTINUATION.md, CHANGELOG/VERSION at release.

Data flow — tmx new -s NAME

1. Resolve NAME (existing sanitize/color-parse logic, unchanged).
2. has-password NAME → 0 / 1 / 2.
3. Exit 0 (password exists — live or recycled-dead): masked prompt "Session is password-protected. Enter password:" → verify → **wrong** → **print error, exit 1, no tmux session touched**; correct → proceed silently, no further password prompts.
4. Exit 1 or 2 (no password currently set): proceed to existing collision-check + tmux session create/recreate, unchanged.
5. Only in the exit-1-or-2 case, after the session is confirmed up: masked "Enter password for session (blank = none):" → blank → done, no password. Non-blank → masked "Confirm password:" → match → set-password → attach. Mismatch → "passwords did not match, try again" → retry (max 3) → 3rd failure → kill the session just created, print error, exit 1.

Data flow — tmx attach -t NAME

has-session -t NAME first. Not live → tmx: no session named "NAME" to stderr, exit 1, **no password prompt at all**. Live → existing has-password → verify-once logic (unchanged) → exec attach.

Data flow — wizard (tmx-shell-init.sh.template)

```
printf '[tmx] Enter session name to create (blank = choose
existing session): '
read session_name

if [ -z "$session_name" ] || [ "$session_name" = "default" ]:
    list existing sessions (parse `tmx ls` output, names before
first ':')
    if none exist: return 0 (bare shell, unchanged)
    else: print numbered menu + "0) None (leave, bare shell)"
        read choice
        valid 1..N -> exec sh -c 'exec tmx attach -t "$1" ...
"$chosen_name"
        "0)/blank/invalid-after-retry -> return 0 (bare shell)
else:
    validate charset/length (existing rule: [A-Za-z0-9_.-:#{
{1,80})
    suffix = awk-generated 4 random digits (unless
TMX_EXACT_NAME=1)
    real_name = "${session_name}-${suffix}" (or literal if opted
out)
    exec sh -c 'exec tmx new -s "$1" tmx-shell-init "$real_name"
```

Masked password reader (bash, tmx.template only)

Reads one character at a time via `read -rsn1` from `/dev/tty`, echoes
* per printable character, handles backspace/DEL (erase one * +
one buffered char), terminates on Enter, returns the full string on
stdout. Used at all three password-prompt call sites.

has-password Go contract

```
tmx-state has-password <session>
    exit 0 – record exists AND PasswordHash != ""
    exit 1 – record exists AND PasswordHash == ""
    exit 2 – no record for this session
    (no stdout – pure exit-code contract, mirrors verify-password's
shape)
```

Replaces the double-call-verify-password-with-empty-guess trick
currently in the attach verb — one primitive answers "does this
session have a password" instead of two implementations of the
same question.

Error handling & edge cases

- Wrong password on verify (either verb): error + exit 1, no session touched. No retry loop (security gate, not typo-recovery).
- Confirmation mismatch on brand-new password: 3 retries, then abort + tear down the just-created session if one was made.
- attach on a name with no live session: clean error, no /dev/tty interaction, exit 1.
- Wizard, zero existing sessions on blank input: skip menu, bare shell.
- Wizard, invalid selection: one re-prompt, then bare shell.
- Wizard, suffix collision (~1/10000): surfaced as today's existing "already active" error — no silent retry-with-new-suffix (honestly documented residual gap, not hidden).
- Ctrl-D/EOF during masked read: treated as empty input (matches current read EOF behavior).
- TMX_EXACT_NAME + genuine name collision: unchanged existing collision-refusal (exit 4).

Test plan

- **Unit (Go):** TestHasPassword — 0/1/2 cases, existing set/verify tests untouched.
- **77_password_masked_echo.sh** — PTY-driven, asserts * echo + backspace, never plaintext.
- **78_wizard_suffix_appended.sh** — PTY-driven, asserts created name matches ^name-[0-9]{4}\$, two runs with the same base name produce different suffixes.
- **79_tmx_exact_name_optout.sh** — TMX_EXACT_NAME=1 → literal name, no suffix.
- **80_wizard_select_existing.sh** — PTY-driven, menu lists pre-created sessions + "0) None"; selection attaches; "0"/invalid → bare shell.
- **81_wizard_select_password_protected.sh** — same, chosen session is password-protected: exactly one prompt, wrong rejects, correct attaches.
- **82_new_password_confirm_flow.sh** — fresh name: password+confirm double-prompt, match succeeds, mismatch retries + message, 3x mismatch aborts and session is torn down (has-session fails afterward).
- **83_open_existing_password_single_prompt.sh** — root-cause regression guard (§11.4.115 RED-on-broken-artifact): reproduce recycle-then-reopen, assert exactly ONE password prompt, correct attaches, persisted hash unchanged after reopen.
- **84_attach_dead_session_no_prompt.sh** — attach on a dead session with a persisted password: no prompt, clean error, exit 1.

- **66_session_password.sh**: add has-password 0/1/2 cases.
- **68_session_lifecycle.sh**: rewrite C6 (verify-once semantics) and C7 (set+confirm-twice semantics) per §11.4.120 gate reconciliation, citing this design doc in the commit.
- **Challenges bank** (scripts/challenges/tmux.yaml): new entries for suffix-on-create, single-prompt-reopen, double-prompt-create, wizard-picker-on-blank — each scored PASS only on captured PTY evidence.
- **HelixQA**: no HelixQA submodule is currently wired into this project (confirmed absent during exploration) — honest gap, tracked as a follow-up item rather than fabricated.
- **Meta-test (§1.1)**: paired mutation per new gate (e.g. mutate the masking function to echo plaintext → test 77 must FAIL; mutate has-password to always return 2 → test 83 must FAIL).

Documentation plan

Doc	Update
README.md	Commands table note on new password/wizard semantics.
docs/guides/tmx-shell-integration.md	Rewrite wizard section: suffix, picker, masking, TMX_EXACT_NAME.
docs/guides/tmx-session-passwords.md (new)	Standalone password-subsystem guide: create-flow, open-flow, masking, recycle interaction, delete-reset semantics.
FAQ	New Q&As: double-password-prompt history, exact-name scripting, existing-session picker.
Diagrams	Wizard decision tree + new/attach password decision flow (Mermaid, rendered as images per §11.4.168).
Workable-items DB	4 new ATM-NNN tickets (suffix, masking, double-prompt bug, wizard picker), full descriptions, closure evidence on Fixed.md migration.
docs/CONTINUATION.md	Live state, same commit as code.
CHANGELOG.md + VERSION	Bumped at release phase.

Out of scope

- No change to `tmx ls`, `tmx kill-session`, `tmx delete`, `tmx reload` verb contracts beyond what's described above.
- No change to color parsing, cwd-restore, recycler timing, or OS-native isolation (systemd scope / rlimit wrapper) mechanics.

- No HelixQA autonomous session (submodule not present in this project — tracked as a follow-up, not fabricated).